

RAM SASIDHAR PUTCHA

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EDUCATION

Integrated M.Tech, Computer Science & Engineering Vellore Institute of Technology, Chennai	2022 – 2027 CGPA: 8.60 / 10
Senior Secondary (XII), CBSE Chettinad Vidyashram, Chennai	2022 85.20%
Secondary (X), CBSE Hindu Sr Sec School, Indira Nagar, Chennai	2020 93%

EXPERIENCE

Machine Learning Intern Vizag Steel Plant - RINL, Visakhapatnam	Jun 2025 – Jul 2025
- Designed and deployed a PPE detection system using YOLOv8, Python and OpenCV for live gate surveillance. - Integrated inference pipeline with Node.js backend and Firebase for real-time compliance monitoring. - Achieved 91% detection accuracy; enabled automated safety reporting. - Delivered a production-ready solution independently; received Letter of Recommendation.	
UI/UX Design Intern 3F (Virtual)	Jul 2024 – Oct 2024
- Designed and prototyped mobile interfaces using Figma, improving usability through iterative testing. - Integrated AI-generated assets to accelerate design workflows. - Optimized data conversion pipelines to improve cross-team collaboration.	
AI-ML Intern Winfo Solutions Private Limited, Hyderabad	May 2024 – Jul 2024
- Curated and processed 500+ invoice documents for model training. - Improved extraction accuracy from 50% to 85% using Google Cloud Document AI. - Implemented preprocessing and validation routines to enhance model robustness.	

PROJECTS

LLM-Based Emotional Analysis of Live YouTube Comments for Real-Time Streaming Analytics

Engineered a streaming data analytics pipeline integrating APIs and large language models for continuous inference on live data streams. Optimized latency and processing workflows for scalable real-time analytics applications and visualization systems.

IoT-Enabled Smart Safety Helmet with Embedded Wear Detection and Real-Time Fall Monitoring

Designed and implemented an embedded IoT safety system integrating capacitive, proximity, and IMU sensors for real-time wear-state verification and motion analysis. Developed microcontroller-based sensing and event-detection logic enabling orientation tracking and fall detection under dynamic operating conditions. Implemented low-power peer-to-peer wireless communication using ESP-NOW protocol for reliable alert transmission in infrastructure-constrained environments. Optimized sensor data acquisition and interrupt-driven processing workflows for low-latency edge decision making in safety-critical applications.

A Multi-Modal Framework for Road Condition Monitoring Using Dashcam Imagery, Inertial Sensing, and Geospatial Event Detection

Accepted at the 16th International IEEE Conference on Computing, Communication and Networking Technologies (2025).

Developed a real-time monitoring framework integrating computer vision and inertial sensing for anomaly detection using multi-modal data fusion. Designed an Android-based crowdsourced data collection system enabling continuous sensing, geospatial tagging, and cloud-based analytics using Firebase. Improved detection reliability through IMU-based jerk analysis supporting data-driven safety monitoring and large-scale analytics workflows.

Intelligent OBD Vehicle Tracker – Connected Diagnostics and Real-Time Telemetry Platform

Designed a connected diagnostics system interfacing with vehicle ECUs via Bluetooth-enabled OBD-II modules for live telemetry acquisition. Implemented real-time sensor data streaming and lifecycle monitoring supporting predictive maintenance analysis. Developed offline-capable mobile architecture with synchronized cloud storage ensuring reliable system performance under constrained connectivity environments.

TECHNICAL SKILLS

Programming	Python, C, C++, Java, JavaScript, SQL, Bash, R, HTML, CSS
Frameworks/Libraries	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, Pandas, NumPy, Streamlit
Databases	MySQL, Neo4j, Firebase
Tools	Git, Databricks, Arduino/ESP32 Toolchain, Google Cloud (Document AI), Figma
Data Visualization	Power BI, Tableau
Concepts	Computer Vision, Deep Learning, NLP, EDA, Transfer Learning, Embedded IoT

CERTIFICATIONS

Programming with JavaScript (Coursera)	Jan 2025 – Mar 2025
Introduction to Front-End Development (Meta)	Dec 2024 – Jan 2025
Quantum Machine Learning (KwantamG Research Labs)	Jul 2024 – Aug 2024
Machine Learning A-Z: AI, Python R (Udemy)	May 2024 – Oct 2024

EXTRACURRICULAR ACTIVITIES

- District Youth Coordinator and volunteer for Sri Sathya Sai Organization; organized community service initiatives, such as food and clothing donation drives.
- Award recipient in Karate competitions with advanced belt certifications.
- Multiple-time school-level winner in Bhagavad Gita recitation competitions.